

Mathematics

A Cultural Heritage

Lecture 2

Numbers – Part 1

Numbers and Abstraction

Brief Review of last time

1. Not “normal” math class
2. Meta-speech about math (not under the hood)
3. Reading, writing, explaining key course material

Course Topics

*Funny Side of
Mathematics*

Mathematics,
literatures, movies,
and more

*Mathematics
and the Arts*

Mathematics
and Education

Two mathematics
cultures and a few
mathematical lives

Numbers

Numbers
at work

Our Obsessions
with Numbers

**Mathematics
and Society**

Mathematics and Truth
(Philosophical Aspects
of Mathematics)

**THEOREMS (PROBLEMS)
AND PROOFS**

Lecture Outline

1. What are numbers? Ways to represent numbers:

- Counting/Measuring/Tallying
- Written numbers throughout history
- Number bases

2. Other kinds of numbers

3. What is Mathematics?

- Abstraction
- Operations on numbers

Numbers in Everyday Life

We see numbers every day



From: <http://www.englishexercises.org/makeagame/viewgame.asp?id=8254>

Your UID: u3518...

Course code: CCST9037

Task	Weighting
In-class worksheets	10%
Tutorial participation	20%
Written assignments	30%
Essay & presentation	40%

Last Week's Reading Assignment

URL: www.economist.com/node/12847128

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Numbers

Easy as 1, 2, 3

[Comment \(19\)](#) [E-mail](#)

People come into the world ready to count its wonders

Dec 30th 2008 | From the print edition

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THE baby is just one day old and has not yet left hospital. She is quiet but alert. Twenty centimetres from her face



Illustration by Jon Berkley

researchers have placed a white card with two black spots on it. She stares at it intently. A researcher removes the card and replaces it by another, this time with the spots differently spaced. As the cards alternate, her gaze starts to wander—until a third, with three black spots, is presented. Her gaze returns: she looks at it for twice as long as she

Advertisement



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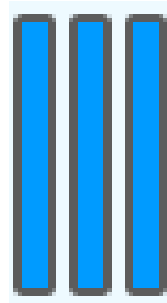
Reminder

Read
the
reading
assignments!

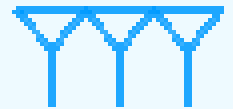
What is a Number?

1. What is the meaning of, say, “three”?

3



three



tres



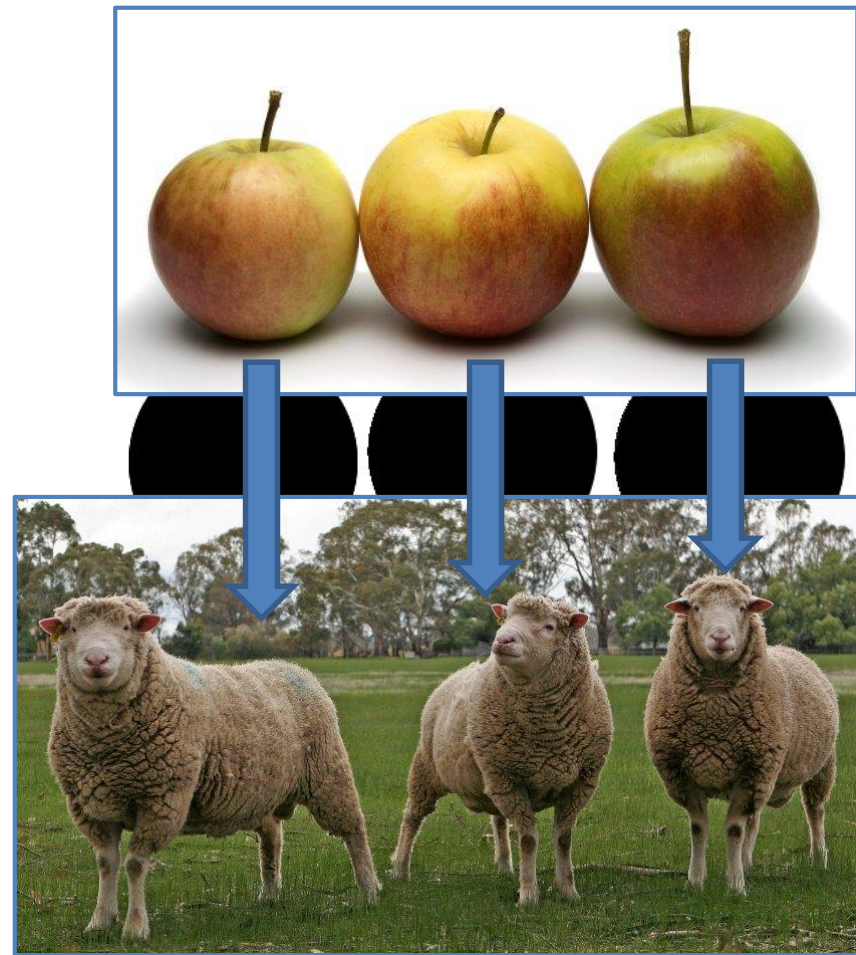
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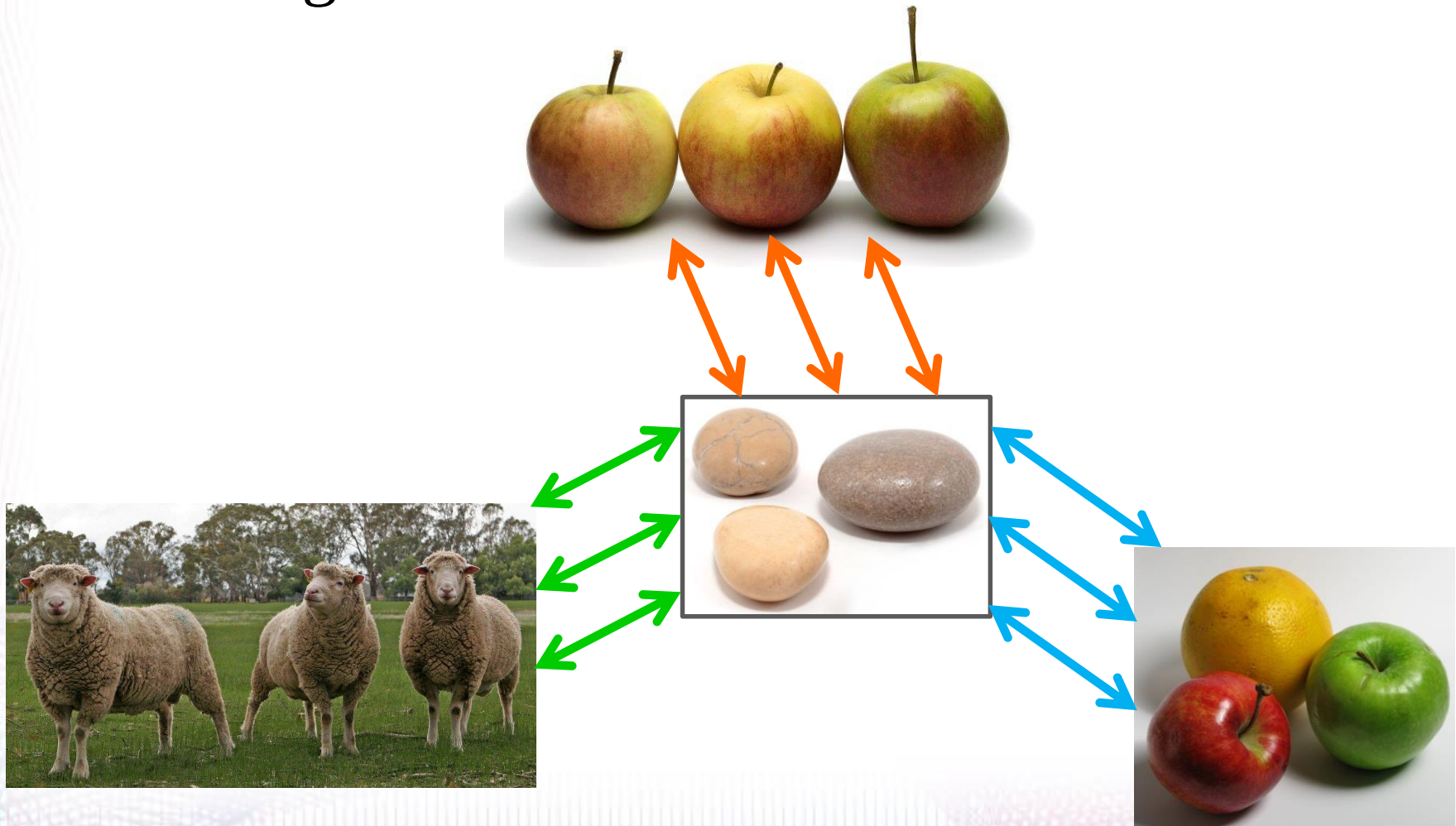
Numbers and Correspondence

- Three apples?
- Three sheep?
- What makes them the same?



Pairing Objects

- Matching one-to-one



Tally Marks



Larry	I
Bobby	
Tony	II
Linda	



正 正 正 正 正 正

The Way of Mathematics

1. The “**Way**” of Mathematics:

- **Generalization** (of particular cases)
- **Abstraction** (of concrete cases)
- ...

Abstraction and Mathematical Worlds

- Idea: Move question to an easier setting
- Examples:
 - Numbers/Correspondence/Tallying
 - ...

The Ishango Bone

- More than 20000 years old
- Division/Multiplication?
- Odd numbers?

19 + 17 + 13 + 11 = 60



 7 + 5 + 4 + 1? + 1 + 9 + 8 + 4 + 6 + 3 = 48

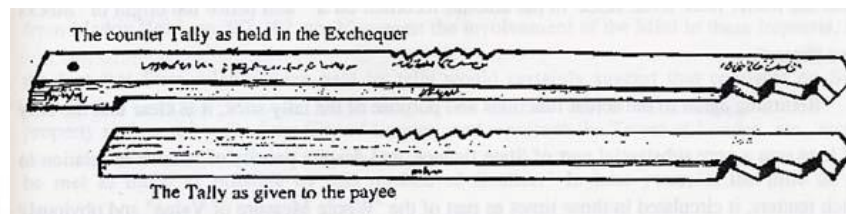
9 + 19 + 21 + 11 = 60



Counting and tallying

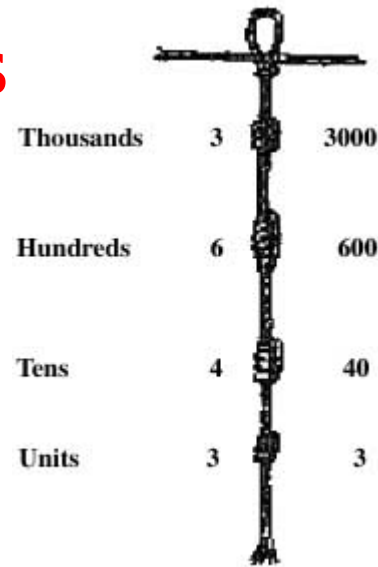
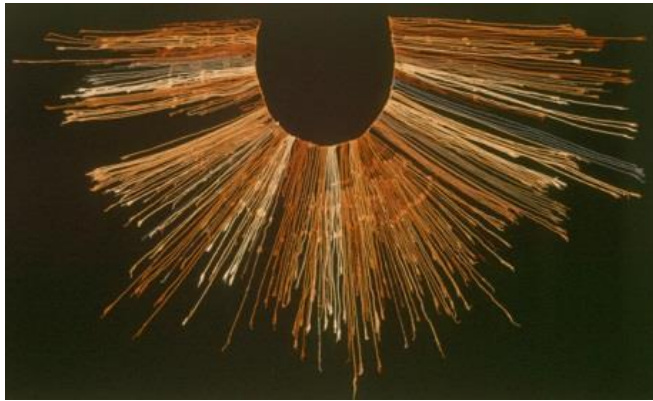
1. Tally sticks

- Single Tally
- Split Tally

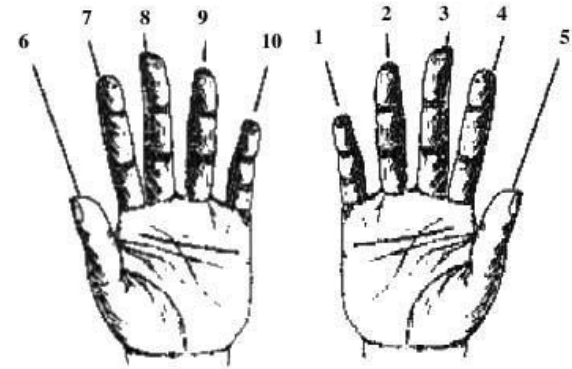


Other ways to count

- Quipu used by Incans



- Body



- Prayer Beads (little effort)



"Buddhist rosary 01" by ja:利用者:
©CC BY-SA 3.0 via Wikimedia Commons



Fingers – Cultural differences



Still of Michael Fassbender and Diane Kruger
in Inglourious Basterds (2009)
©Universal Studios



Stills of Diane Kruger
in Inglourious Basterds (2009)
©Universal Studios

Finger – Cultural differences

- What number is this?
 - 7 (here)
 - 8 (most of mainland)
 - 2 (Germany)



One, Two, Many

Video excerpt from **The Story of 1**



Summary

1. Counting/Tallying

2. Counting Devices

3. Number Systems

- Egyptian
- Babylonian
- Greek
- Roman
- Chinese
- Hindu-Arabic
- Positional Systems/Number zero
- Number Bases
- Societies without Counting Words